

GRB 090804

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ABSTRACT

We present the complete timing and time-resolved spectral analysis of the single-pulse burst GRB 090804 (*Fermi*/GBM trigger bn090804940), the seventh burst of a 106-burst single-pulse campaign run on a staged, gate-reviewed pipeline. The burst is the campaign’s most compact ($T_{90} = 6.68^{+0.11}_{-0.15}$ s), a clean FRED ($\varphi = 0.232 \pm 0.015$, $R^2 = 0.98$), and it contributes three campaign firsts. First, its spectral lag is consistent with zero: $\tau = 0.037^{+0.034}_{-0.030} \pm 0.098_{\text{win}}$ s (25–50 vs. 100–300 keV), with the scan members changing sign across fit windows — the first burst of the sample without a measurable lag, at high cross-correlation significance (49σ). Second, its *time-integrated* spectrum is won by a blackbody-bearing model — Band+BB with $kT = 25.1 \pm 1.4$ keV — the campaign’s first thermal integrated winner, and the temperature stays in a narrow band: the two BB-winning blocks give $kT = 20.2 \pm 1.1$ and 25.4 ± 1.2 keV (a 25% spread, though the two differ at 3.0σ), in direct contrast to burst #4’s factor-4 cooling (the Wien peaks of all BB-*winning* fits edge-clean; the integrated margin over the best BB-free model is $\Delta\text{AIC} = 2.3$, a near-tie we report as a strong candidate, not a detection). Third, the wavelet variability estimator lands on a third distinct grid rung (181 ± 14 ms, after 215 and 256 ms in earlier bursts), mapping the estimator’s discrete scale ladder; the canonical windowed MVT is 24.3 ± 2.8 ms and Haar gives 453 ± 42 ms. In the uniform Band convention the low-energy index sits above the synchrotron line of death through nearly the whole burst ($\alpha = -0.10$ to -0.33 across the first five blocks) and crosses only in the final block (-1.00) — the fourth consecutive epoch-dependent violation, here with the crossing displaced to the very end. The 24×8 grid renders under the no-model-dropped rule with one labeled engine-refusal cell; b_0 ’s effective-area factor rails at 0.80 and is disclosed.

Keywords: Gamma-ray bursts (629) — Time domain astronomy (2109)

1. INTRODUCTION

GRB 090804 triggered the *Fermi* Gamma-ray Burst Monitor (Meegan et al. 2009) on 2009 August 4. As burst #7 of the single-pulse campaign it runs on the pipeline frozen from bursts #1–#2, and it anchors the thermal end of the campaign’s empirical axes: where burst #4 showed a cooling photospheric track and bursts #5–#6 showed tie-level candidates only, GRB 090804 pairs a thermal integrated winner with a stable ~ 20 –25 keV temperature — the configuration closest to a persistent photosphere the sample has yet produced. Its zero-consistent lag simultaneously extends the campaign’s lag-width sequence to the origin.

2. OBSERVATIONS, DETECTORS, AND APPROVED SELECTIONS

We use TTE data from NaI detectors n_3 , n_4 , n_5 and BGO b_0 (Figure 1), with the spectral convention of Chand et al. (2020): NaI 8–900 keV excluding the 30–40 keV K-edge, BGO 0.2–38 MeV. Background and

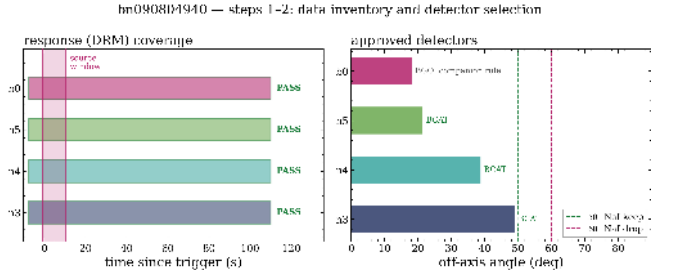


Figure 1. Detector selection for GRB 090804: source angles with the 50° rule; selected detectors n_3 , n_4 , n_5 , b_0 .

source selections are the human-approved Stage-1 decisions (source window $(-0.91, 10.33)$ s; Figures 2–3). Effective-area factors float in $[0.8, 1.2]$; the integrated winner carries $n_4 = 1.00$, $n_5 = 0.95$, and b_0 railed at 0.80 — the recurrent BGO-side rail, disclosed as a standing caveat on every spectral statement.

3. TIMING ANALYSIS

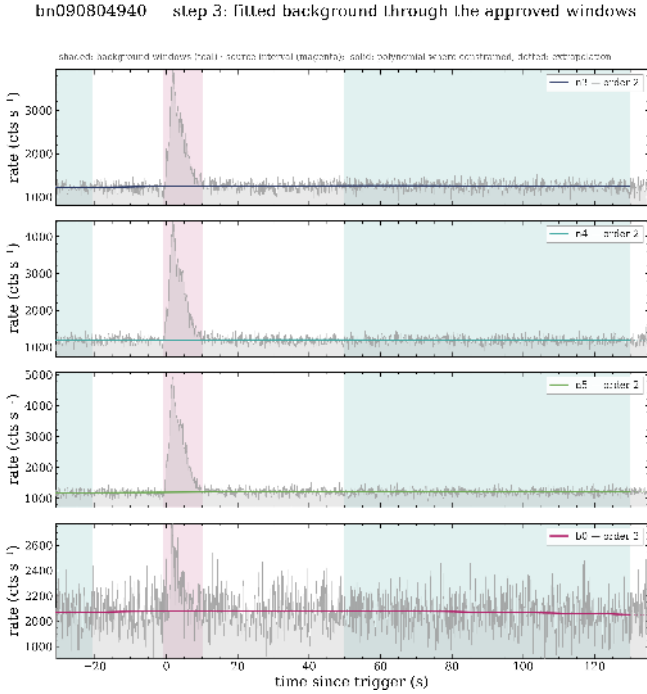


Figure 2. Approved background intervals and per-detector polynomial fits.

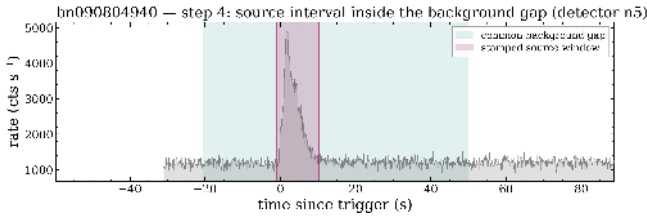


Figure 3. Approved source interval over the background-subtracted light curve.

All timing runs on the approved selections, reference NaI n_5 .

3.1. Durations

The windowed duration is $T_{90} = 6.68^{+0.11}_{-0.15}$ s spanning [0.48, 7.16] s (covariance-true Monte Carlo; window not truncated) — the campaign’s most compact burst — with $T_{50} = 2.86$ s ($T_{50}/T_{90} = 0.43$).

3.2. Pulse Shape

The Norris et al. (1996) profile wins the shape comparison, and the two-sigmoid of Gowri et al. (2025) gives $\varphi = s_l/s_r = 0.232 \pm 0.015$ ($R^2 = 0.98$, the campaign’s best-fit pulse): a clean FRED, returning the sample to the FRED family after two consecutive mixed pulses (Figure 4). The shape axis now reads $\varphi = 0.017$ –0.41 across seven bursts.

3.3. Minimum Variability Timescale

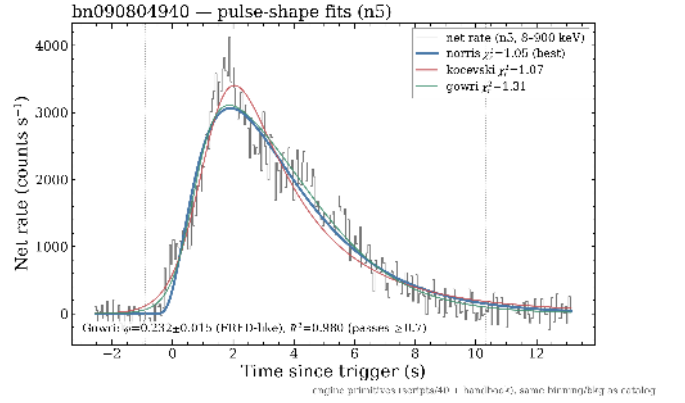


Figure 4. Pulse-shape fits to the net light curve; Norris wins; $\varphi = 0.232 \pm 0.015$, a clean FRED.

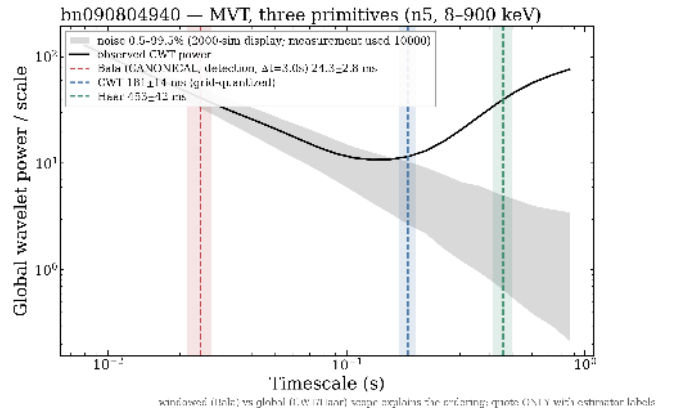


Figure 5. Minimum-variability estimators with explicit labels: canonical windowed 24.3 ± 2.8 ms; CWT 181 ± 14 ms (third distinct grid rung); Haar 453 ± 42 ms.

All three estimators return measurements (Figure 5): the windowed framework of Bala et al. (2026) — the canonical value — finds 24.3 ± 2.8 ms ($\Delta t = 3$ s segment, detection); the global CWT (Vianello et al. 2018) returns 181 ± 14 ms; the Haar cross-check (Golkhou & Butler 2014) gives 453 ± 42 ms. The CWT value is the estimator’s *third* distinct rung across the campaign (181, 215, 256 ms; spacing ratio $\simeq 1.19$; the 215 ms rung is occupied by bursts #1, #2, #3, #5): the discrete scale ladder is now mapped at three points, and cross-burst CWT comparisons are quantized at the $\sim 20\%$ level by construction — the campaign quotes CWT only with this label.

3.4. Spectral Lag: Consistent with Zero

Scanning the cross-correlation fit window around the peak (half-widths 0.7–2.8 s; Figure 6) gives $\tau = 0.037^{+0.034}_{-0.030} \pm 0.098_{\text{win}}$ s (25–50 keV vs. 100–300 keV; detector n_5 , 16 ms resolution, peak significance 49σ). The scan members are 0.143, 0.037, -0.013 , -0.053 s — they *change sign* across fit windows, and the window

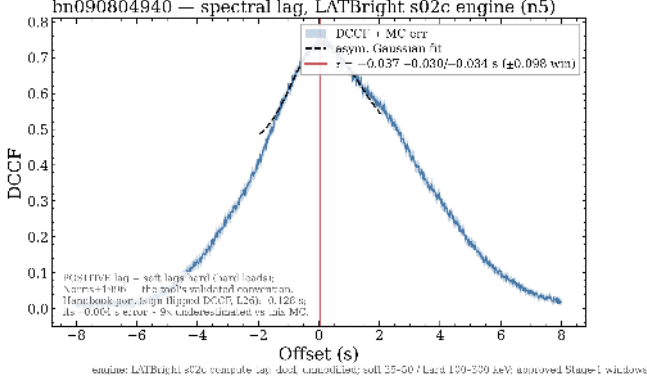


Figure 6. Cross-correlation lag with the window-scan systematic: the scan members change sign and the systematic dominates — consistent with zero.

systematic (± 0.098 s) exceeds the statistical error: the lag is consistent with zero. This is the campaign’s first zero-lag burst, and it lands at the origin of the accumulating lag-width plane ($\tau/T_{90} = 0.6\% \pm 1.5\%$, the uncertainty from the window systematic, against 8.5%, 4.2%, 2.6%, 5.4%, 6.0% for bursts #1, #2, #4, #5, #6 — all of whose scan members keep one sign).

3.5. Per-band Durations

Per-band light curves and the $T_{90}(E)$ relation are shown in Figure 7; the population slope of Qin et al. (2013) is drawn for reference only.

4. SPECTRAL BINNING

Seven Bayesian blocks plus the integrated spectrum (Figure 8), from the frozen 3ML Bayesian-block + significance hybrid (blocks computed 2026-08-12, reused unchanged).

5. TIME-RESOLVED AND TIME-INTEGRATED SPECTROSCOPY

5.1. Method

All 24 campaign models fit to every bin (PGstat; AIC ranking (Akaike 1974)); no LLE/LAT data exist for this burst (inclusion gated on data availability, never detection significance). The 24×8 grid renders under the no-model-dropped rule with one labeled engine-refusal cell (BandR+CPL, integrated spectrum); per-bin all-model parameter tables accompany the products.

5.2. Integrated Spectrum: the First Thermal Winner

The integrated spectrum ($-0.05, 10.33$) s is won by Band+BB (Figure 9): Band $\alpha = -0.14 \pm 0.28$, $E_p = 38 \pm 6$ keV, $\beta = -2.25 \pm 0.14$, plus a blackbody at $kT = 25.1 \pm 1.4$ keV (AIC = 5300.5). The runner-up is also BB-bearing (SBPL+BB, $\Delta\text{AIC} = 0.4$), and the best BB-free model (DSBPL) trails by $\Delta\text{AIC} = 2.3$: the thermal

preference is model-robust at the top of the ranking but the margin is a near-tie, so we report the campaign’s *first thermal integrated winner* as a strong candidate, not a detection. The Wien peak $3.92 kT = 98$ keV is far from the 8 keV edge.

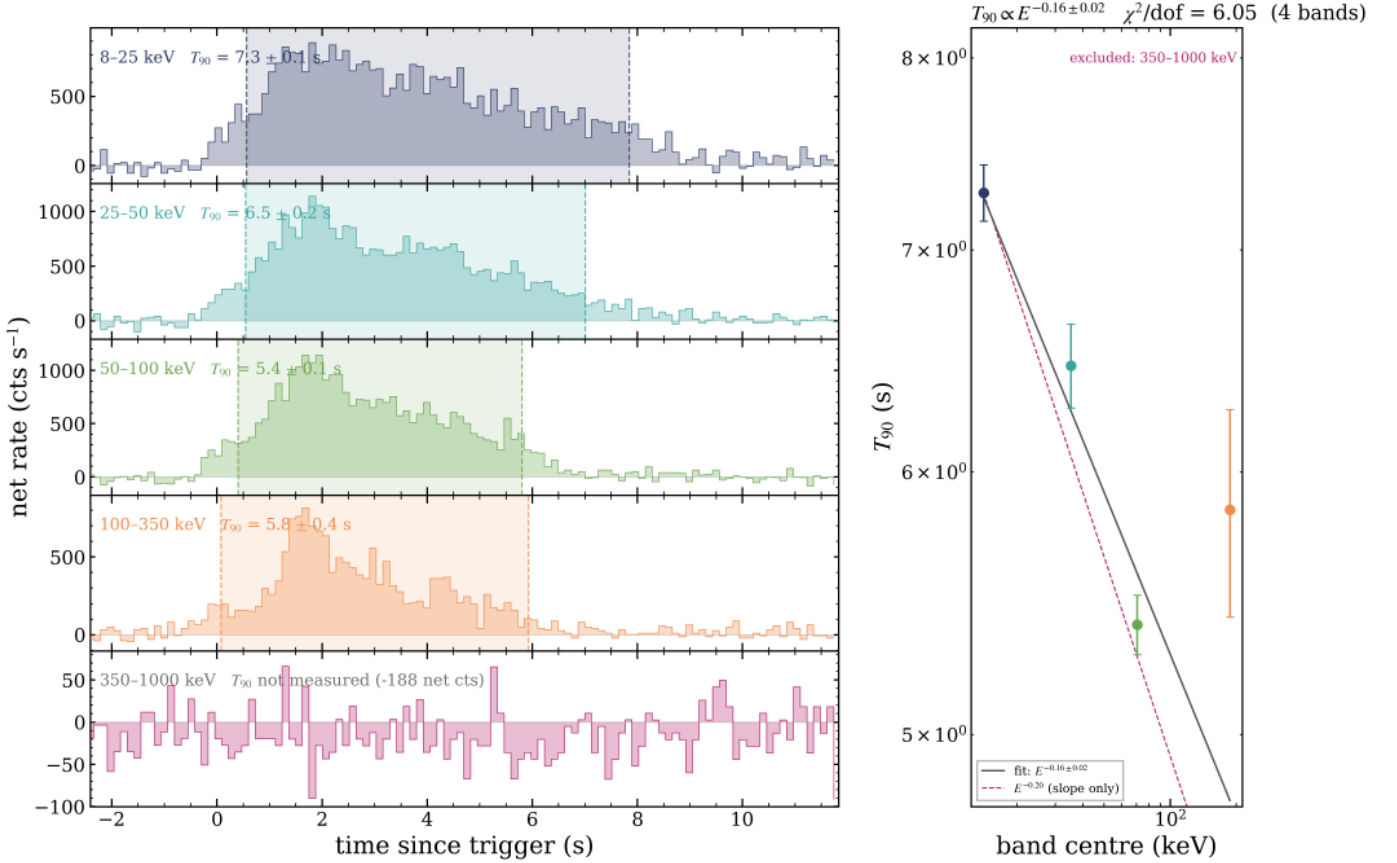
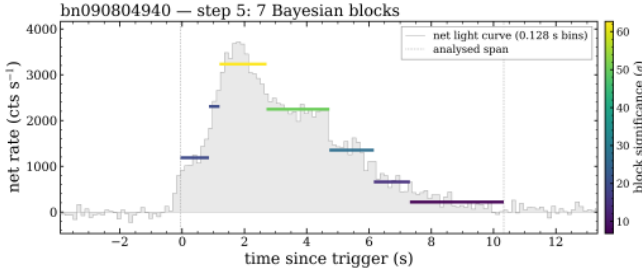
5.3. Time-Resolved Story: a Stable Photospheric Candidate

BB-bearing models win two blocks — block 0 (CPL+BB, $kT = 20.2 \pm 1.1$ keV, margin 1.4 over plain CPL) and block 3 (Band+BB, $kT = 25.4 \pm 1.3$ keV, margin 0.3; Figure 12) — and the three BB temperatures (integrated, block 0, block 3) stay within a narrow ~ 20 – 25 keV band — a 25% spread, though block 0 and block 3 differ at 3.0σ , so this is a *quasi-stable* track rather than strict isothermality — in direct contrast to burst #4’s monotonic $22 \rightarrow 6$ keV cooling. The two configurations bracket the photospheric phenomenology the campaign is accumulating: cooling tracks and quasi-stable temperatures both occur in single-pulse bursts (Figure 10). Separately, the step-9 nested-LRT gate flags significant *cool* blackbodies in blocks 2 and 6 ($kT = 6.0$ and 4.5 keV, Wien peaks 23.6 and 17.5 keV — edge-marginal and edge-constrained respectively), which the AIC ranking does not promote; per the campaign’s edge protocol these are recorded as edge-zone candidates, not detections.

In the uniform Band convention α runs $-0.10, -0.13, -0.37, -0.44, -0.33$ across blocks 0–4 — all above the $-2/3$ line of death (the block-4 Band fit is itself β -railed and flagged invalid; its α is quoted with that caveat) — and crosses only in the final block (-1.00 , with E_p at the engine’s 30 keV floor, that fit bound-railed). Block 5’s uniform-Band read ($\alpha = +0.90$) sits in a $\Delta\text{AIC} = 0.07$ tie-cluster whose winners are composite models; we treat that single-Band value as unreliable in a composite-dominated bin and exclude it from the trend statement. The violation-then-crossing pattern is the campaign’s fourth consecutive epoch-dependent case, here with the crossing displaced to the burst’s very end. E_p declines from ≈ 83 – 85 to 30 keV with one block-2 excursion to 111 keV.

6. QUALITY CONTROL AND PROVENANCE

Step-9 QC (Figure 13) is built from the canonical campaign fit table; every figure carries its producing script’s SHA-256 in a JSON sidecar; numbers are recomputed from product files; all figures pass a fresh-context vision gate before delivery, enforced by a pre-delivery hook. Fits come from the campaign’s one-invocation 24-model engine run with best-minimum-per-(bin, model) canonicalization.

bn090804940 — step 7: energy-resolved T_{90} (detector n5); dashed = t_5 , t_{95} **Figure 7.** Per-band light curves with individual T_{90} spans and the $T_{90}(E)$ relation.**Figure 8.** Bayesian-block spectral bins over the light curve.

7. SUMMARY

1. GRB 090804 is the campaign's most compact burst ($T_{90} = 6.68^{+0.11}_{-0.15}$ s), a clean FRED ($\varphi = 0.232 \pm 0.015$, $R^2 = 0.98$).
2. Its lag is the campaign's first consistent-with-zero measurement: $\tau = 0.037^{+0.034}_{-0.030} \pm 0.098_{\text{win}}$ s, with sign-changing scan members — the lag-width plane now extends to the origin.
3. The integrated spectrum is the campaign's first BB-bearing winner (Band+BB, $kT = 25.1 \pm$

1.4 keV; best BB-free model at $\Delta\text{AIC} = 2.3$) — a strong candidate, not a detection.

4. The blackbody temperature stays in a narrow band (20.2, 25.4, 25.1 keV in block 0, block 3, integrated; 25% spread, block 0 vs 3 at 3.0σ) — the campaign's first quasi-stable photospheric track, the counterpart to burst #4's cooling track.
5. The CWT variability estimator lands on its third distinct grid rung (181 ms; ladder now 181/215/256, ratio $\simeq 1.19$); the canonical MVT is 24.3 ± 2.8 ms.
6. The synchrotron line of death is violated through nearly the whole burst ($\alpha = -0.10$ to -0.33 , blocks 0–4) with the crossing displaced to the final block (-1.00 ; that fit E_p -railed at the 30 keV floor) — the fourth consecutive epoch-dependent violation.
7. Standing caveats: b_0 rails at 0.80; block 5's uniform-Band read excluded as composite-contaminated; one labeled engine-refusal cell

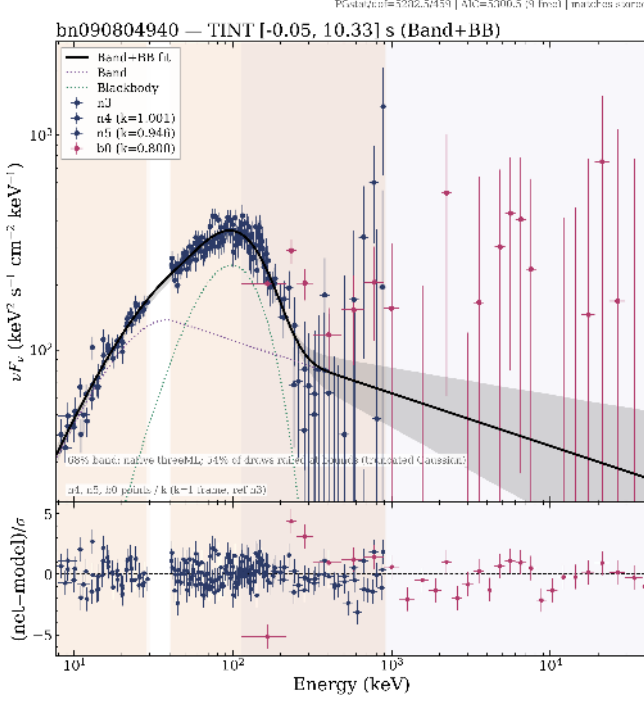


Figure 9. Integrated-spectrum winner: Band+BB with $kT = 25.1$ keV (components dotted) — the campaign’s first BB-bearing integrated winner. 68% band from propagated posterior draws; PGstat/dof, AIC, and EACs stamped on-figure.

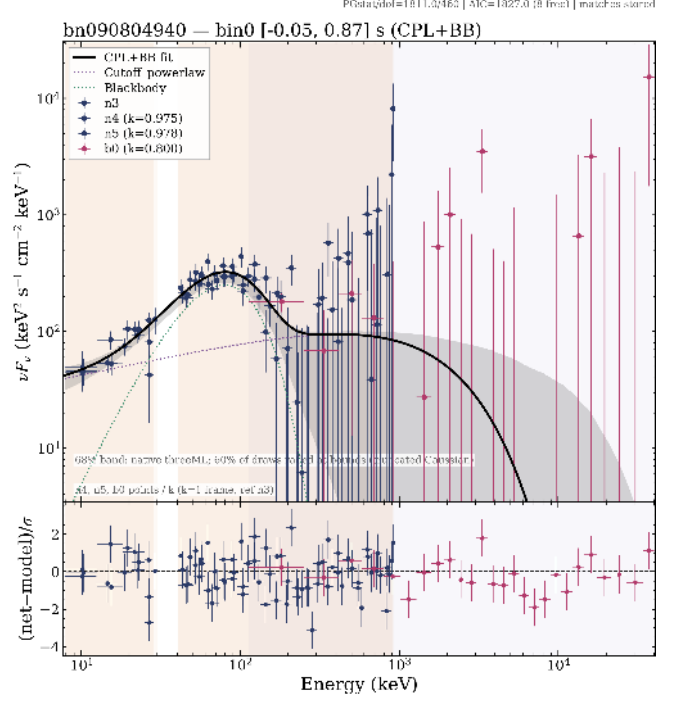


Figure 11. Block 0: CPL+BB winner ($kT = 20.2$ keV, components dotted), margin $\Delta AIC = 1.4$ over plain CPL.

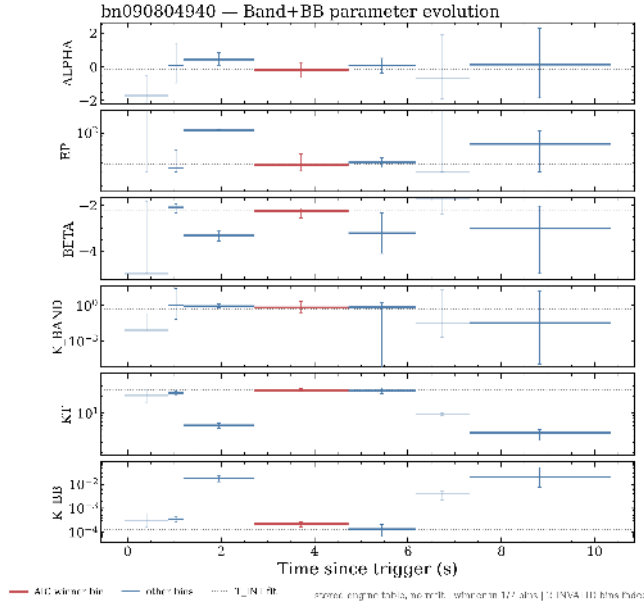


Figure 10. Band+BB parameter evolution: the blackbody temperature is stable at ~ 20 – 25 keV where the model wins — the campaign’s first isothermal photospheric track.

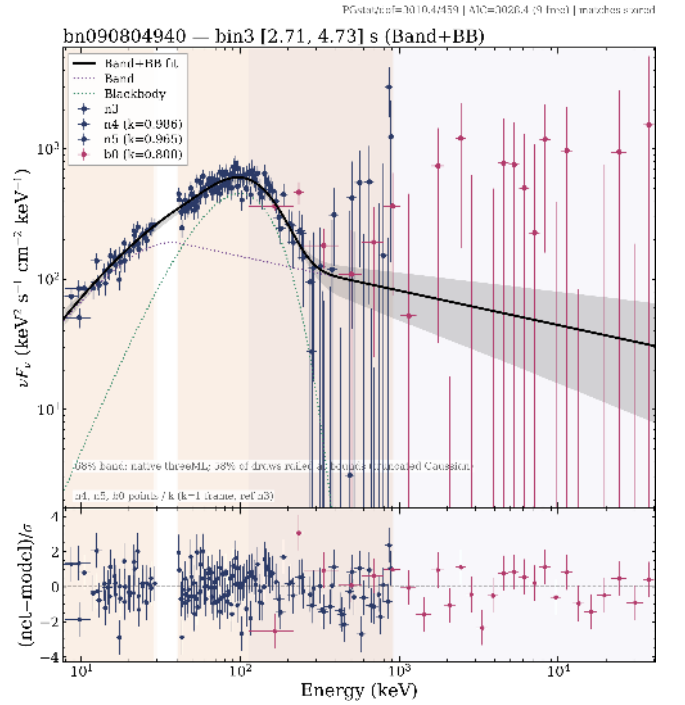


Figure 12. Block 3: Band+BB tie winner ($kT = 25.4$ keV) — the temperature matches block 0 and the integrated fit.

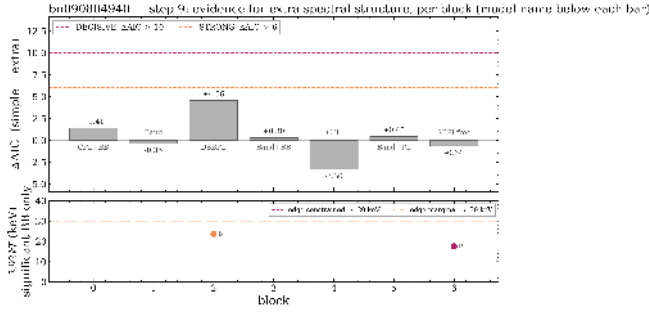


Figure 13. Step-9 scorecard: per-block evidence for extra spectral structure (top) and the 3.92 kT Wien-peak edge check (bottom).

(BandR+CPL, integrated); all numbers provisional pending the PI's campaign review.

APPENDIX

A. ALL-MODEL MONTAGES

Complete 24-model montages for the integrated spectrum and all seven blocks; the winner panel is marked in each; the engine-refusal cell is labeled.

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bn090804940 — TINT [-0.05, 10.33] s — 24 models, AIC-ordered — winner BANDBB

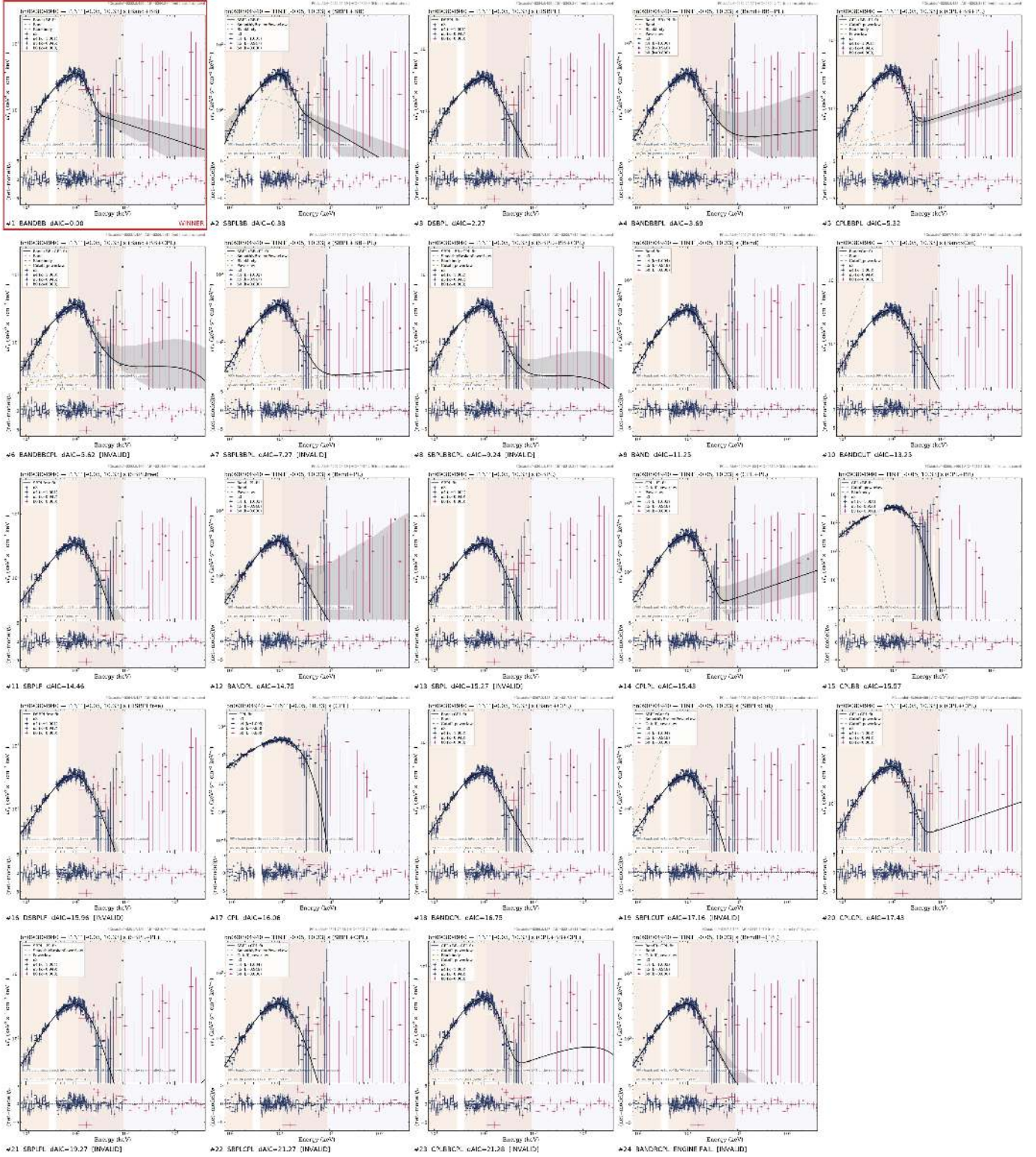


Figure 14. Integrated spectrum: all 24 models (BandR+CPL: labeled refusal cell).

bn090804940 — bin0 [-0.05, 0.87] s — 24 models, AIC-ordered — winner CPLBB

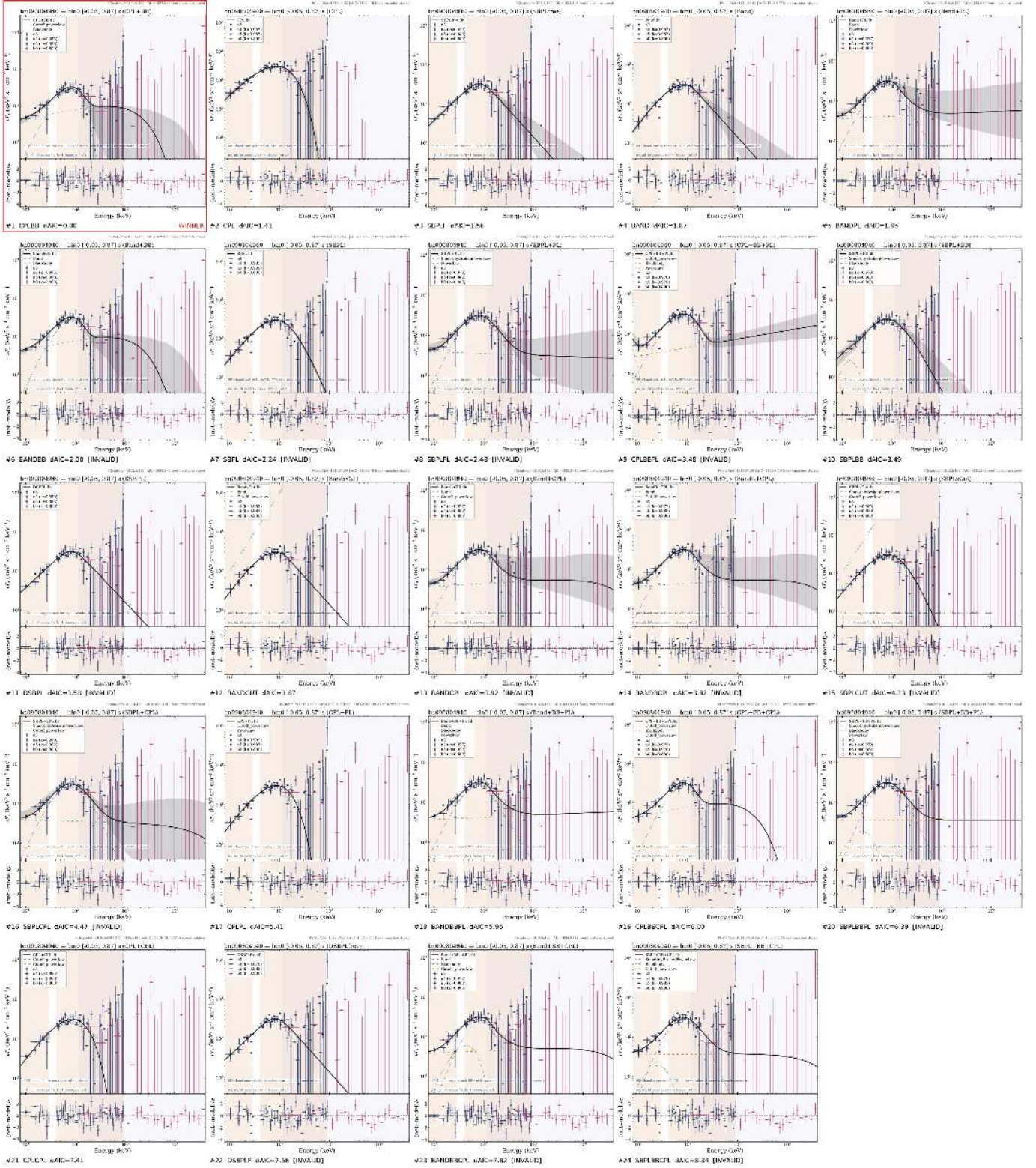


Figure 15. Block 0: all 24 models.

bn090804940 — bin1 [0.87, 1.20] s — 24 models, AIC-ordered — winner BAND



Figure 16. Block 1: all 24 models.

bn090804940 — bin2 [1.20, 2.71] s — 24 models, AIC-ordered — winner DSBPL



Figure 17. Block 2: all 24 models.

bn090804940 — bin3 [2.71, 4.73] s — 24 models, AIC-ordered — winner BANDBB



Figure 18. Block 3: all 24 models.

Figure 19. Block 4: all 24 models.

bn090804940 — bin5 [6.16, 7.32] s — 24 models, AIC-ordered — winner BANDPL



Figure 20. Block 5: all 24 models.

bn090804940 — bin6 [7.32, 10.33] s — 24 models, AIC-ordered — winner SBPLF

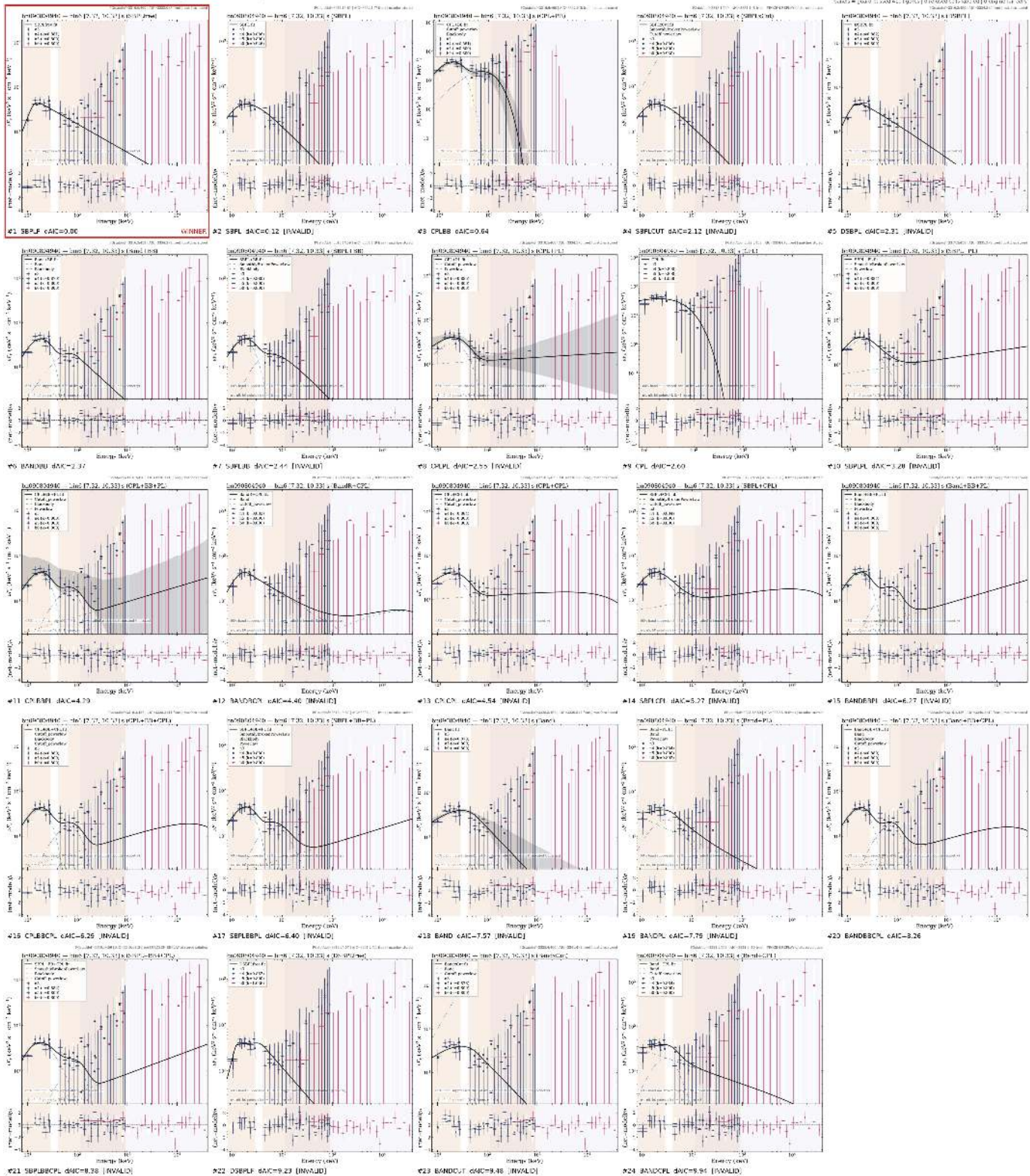


Figure 21. Block 6: all 24 models.